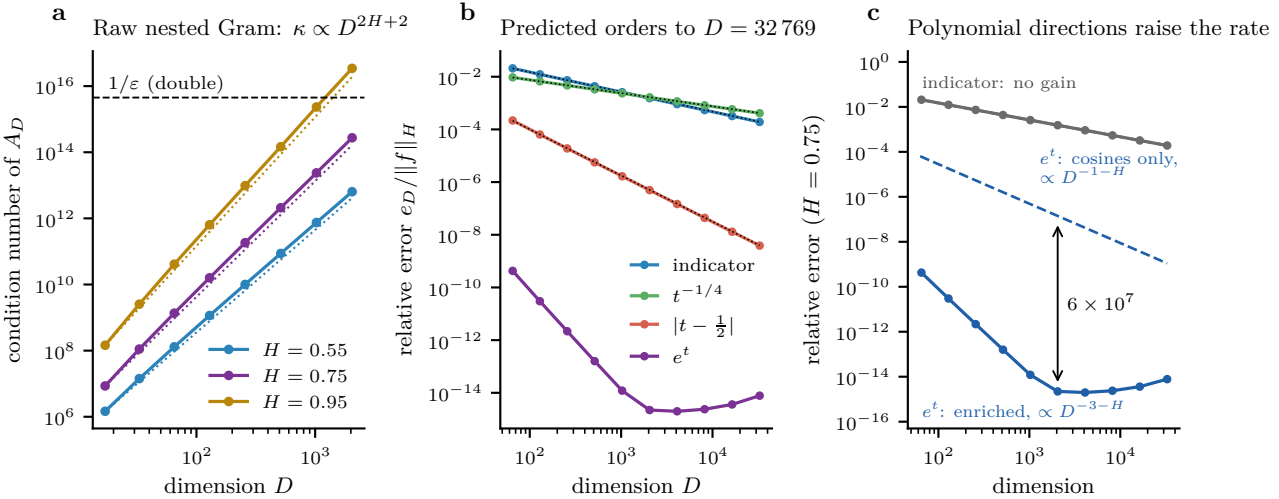

a
Raw nested Gram: $\kappa \propto D^{2H+2}$
$1/\varepsilon$ (double)
condition number of A_D
10^{16}
10^{14}
10^{12}
10^{10}
10^{8}
10^{6}
$H = 0.55$
$H = 0.75$
$H = 0.95$
10^{2}
10^{3}
dimension D
b
Predicted orders to $D = 32\,769$
relative error $e_D/\|f\|_H$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
indicator
$t^{-1/4}$
$|t - \frac{1}{2}|$
e^t
10^{2}
10^{3}
10^{4}
dimension D
c
Polynomial directions raise the rate
relative error ($H = 0.75$)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
indicator: no gain
e^t: cosines only, $\propto D^{-1-H}$
6×10^7
e^t: enriched, $\propto D^{-3-H}$
10^{2}
10^{3}
10^{4}
dimension